

PSN-1: Physics Systems Network with $E(3)$ Equivariant Reasoning and Conservation Law Discovery

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Abstract—We present PSN-1, a neural architecture that learns 3D physical dynamics by combining $E(3)$ -equivariant message passing, multi-head attention reasoning, and conservation law discovery. The model achieves machine-precision rotation equivariance ($< 10^{-6}$) while learning gravitational, elastic, and Lennard-Jones force laws from particle trajectories alone. Ablation studies demonstrate that the equivariant pathway provides stability while the attention reasoning pathway captures complex interaction patterns.

Index Terms—Equivariant neural networks, physics-informed learning, attention mechanisms, conservation laws, graph neural networks

I. INTRODUCTION

Learning physical dynamics from data requires models that respect the symmetries of physics. Two paradigms have emerged: physics-informed neural networks (PINNs) [1] that enforce conservation laws as auxiliary losses, and equivariant networks [2], [3] that build symmetry into the architecture. PSN-1 combines both approaches with a third: *learning* conservation laws from data rather than enforcing them.

II. ARCHITECTURE

Equivariant encoder. Processes rotation-invariant scalar features (mass, speed, kinetic energy, pairwise distances) and rotation-equivariant vector features (centroid-centred positions and velocities) through L layers of message passing. All scalar coefficients are functions of invariant features only, guaranteeing exact $E(3)$ equivariance.

Attention reasoning. Multi-head attention computes interaction weights from scalar features with learnable distance biases. Each head discovers a different interaction type. Force magnitudes are scalar (invariant), and directions are equivariant, preserving overall symmetry.

Conservation discovery. Learns to predict conserved quantities (energy, $\vec{p}$, $\vec{L}$) from particle states with a dual loss: prediction accuracy and constancy across timesteps. This discovers what is conserved without prior knowledge.

Blending. $a_i = g_i \cdot a_i^{\text{equiv}} + (1 - g_i) \cdot a_i^{\text{attn}}$ where g_i is rotation-invariant.

III. EXPERIMENTS

We evaluate on three 3D systems (5 particles each): N -body gravity, spring chains, and Lennard-Jones molecular dynamics, generated with leapfrog integration.

TABLE I

PSN-1 RESULTS. EQUIV = ROTATION EQUIVARIANCE ERROR; DRIFT = RELATIVE ENERGY CHANGE OVER 20-STEP ROLLOUT.

System	MSE	Equiv	Drift	Gate
Gravity	1.3×10^{-12}	1.8×10^{-7}	0.008	0.05
Spring	4.5×10^{-7}	6.6×10^{-7}	0.354	0.24
LJ	4.0×10^{-4}	4.9×10^{-5}	0.227	0.18

All systems achieve machine-precision equivariance. Gravity reaches 10^{-12} MSE. The gate shows the model preferentially uses attention for gravity (all-to-all interactions) and the equivariant pathway for springs (local interactions).

IV. ABLATION

TABLE II
ABLATION ON N-BODY GRAVITY.

Variant	MSE	Equiv	Drift
Full	1.5×10^{-11}	2.0×10^{-7}	0.008
Equiv only	1.8×10^{-10}	1.1×10^{-7}	0.005
Reason only	6.8×10^{-12}	3.2×10^{-7}	0.011
No PINN	1.3×10^{-11}	2.0×10^{-7}	0.007

The equivariant-only variant achieves the lowest drift, confirming its strength for simple systems. The reasoning-only variant achieves the lowest MSE but highest drift, suggesting overfitting to individual timesteps.

V. CONCLUSION

PSN-1 shows that combining equivariance, attention reasoning, and conservation discovery yields accurate, interpretable physics learning. The conservation discovery head opens a path toward discovering unknown physical laws from data.

REFERENCES

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